

Getting Past "But": Finding Opportunity and Making It Happen

by Ruth Malan and Dana Bredemeyer

Innovation is a key to the competitive differentiation equation, and technology factors strongly in that equation. To find differentiating opportunities, we need insight into customer needs, what technology makes possible, and our position in the value network. In this *Executive Report*, we present a story that with charming acuity conveys key principles of innovation, and we relate the lessons to system development. Next, we share pragmatic practices for enhancing the effectiveness of system concept generation, elaboration, and refinement. The process pulses through cycles of divergence (opportunity generation, problem and solution finding) and convergence (decision making). We emphasize tools that foster ideation while creating alignment, balancing exigency with *just enough* design up front to orchestrate the socio-technical process of innovation.

Report

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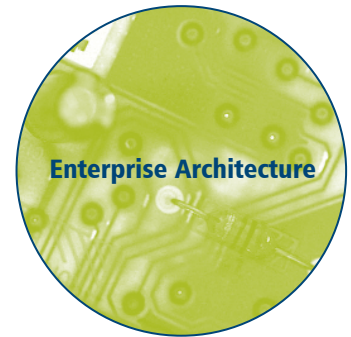
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Getting Past “But”:

Finding Opportunity and Making It Happen

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History supports the position that inertial forces in organizational culture and structure, as well as legacy markets and systems, prevent established companies from seeing or seizing opportunities to reshape the competitive landscape¹ with the result that waves of creative destruction sweep across industries² laying waste to incumbents. Even so, Jim Collins, author of *Good to Great* and *Built to Last*, reminds us that many great companies survive decades of industry-reshaping innovation.³

Of all the innovation traps, the “but” trap is the most insidious: but we aren’t chartered to do that, but that’s too risky, but our customers aren’t asking for that, but that would cannibalize our market. Turf protection, risk aversion, inertia, and cancelled projects mount a personal and organizational history that discourages passionate enthusiasm for finding and running with new ideas. How do we get past the tendency to say “yes, but ...” — the tendency to appeal to all the reasons we can’t — and pursue innovation anyway?

Getting past “but” is about attitude — individual attitude and organizational attitude. Openness to wonder and openness to exploring possibility is where innovation starts. Setting the wheels of change in motion to make it happen involves enrolling more and more people to work in concert on all the contributing innovations that make the overarching innovation real.

Since attitude is fostered by empowerment, we have set out to do our part, sharing what we have learned about setting the innovation process in motion, finding opportunities, and creating the architecture that serves the socio-technical orchestration of contributing innovations.

This *Executive Report*, in key ways, mirrors the innovation process. Like the innovation process, it takes one on a journey: setting motivating context and then pulsing through waves of exploration. In the report, we use a story to introduce key innovation principles and then debrief the lessons. Next, we focus on the

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divergence-convergence cycles of the innovation process, sharing techniques and practices that we have found to be most effective in applying these principles. The first cycle is designed to discover opportunities, which are synthesized in a shared vision. The next cycle expands on the vision, elaborating system capabilities that deliver the value propositions indicated in the vision and elaborating the architecture that will support these capabilities and address dominant challenge and risk. It draws to convergence through scoping and architecture decisions. The cycles keep pulsing, but in this report we have limited ourselves to focusing on the “fuzzy front end” of the innovation process, for it is the front end that is a point of tension in today’s “just do it” world. To help the reader chart a course through this journey, we’ve provided a roadmap in Figure 1.

FINDING OPPORTUNITY: WHY INNOVATION MATTERS

In a world of intense global competition, innovation offers a chance to break away from the pack, to carve out a space where competitive forces are, even if only for a while, damped. It offers a chance to shore up other elements of advantage: identity and customer loyalty, whether due to actual switching costs or a strong position in consumer perception; and a head start on evolutionary adaptation to better serve customers in different use contexts or market segments or to grow the market and gain advantages of scale and leverage across them.

Innovation is seen as the path to organic revenue and profit growth in companies as diverse as IBM, GE, and Procter & Gamble. As Rosabeth Moss Kanter writes: “Innovation is always on the radar, though not necessarily in the top spot. Indeed, innovation comes and goes from the short-list of top strategic objectives.”⁴

In the current cycle of strategic attention, some companies have already passed the span of attention where innovation made it to the short list, many have just gotten there, and others will get there. Strategies shift, in part because cogs are put in place to achieve an objective, bringing other strategic elements center stage. And the competitive context shifts, necessitating a change in priority. Executives have to be strategic about what they make strategic.

The point is, whether innovation is under the top objectives’ spotlight or has fallen off to ninth place, it is a perennial driver of competitive advantage. Being more effective at innovation, and in particular innovation through technology, is of broad concern across industries, not just those we call high tech.

In competitive mature markets, there is always cost pressure. A pure cost strategy is a hard play, and over the long term, cycles through high organizational pain as cost cutting inevitably comes down to people. *Blue Ocean Strategy* looks for new markets, even within old markets.⁵ Innovation through integration of ideas from other domains, invention, and taking customer innovations, concerns, and values into our process all open up new ways to compete — new ways to differentiate our products or services from the low-cost pack.

FINDING OPPORTUNITY: INNOVATION AND TECHNOLOGY

Innovation and technology may not always be headliner concerns of the top executive, but innovation is key to the differentiation equation, and technology factors strongly into that equation. For example, in Cutter’s survey of IT trends for 2008, 35% of respondents said that IT is a key enabler to business innovation.⁶

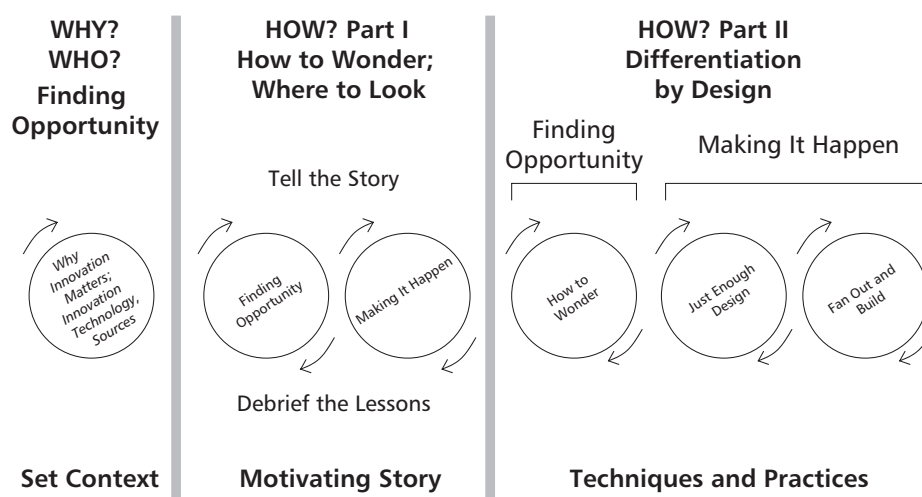


Figure 1 — Report roadmap.

CIOs are seeing their role in an increasingly proactive strategic light. More CIOs are being appointed to boards of directors and more report to the CEO (rather than the CFO or COO), with compensation packages to match their more strategic role.

These changes reflect a growing awareness that information technology and software systems are not just part of the operational substrate but integral to differentiation along any dimension: operational excellence, product leadership, or customer intimacy.

Innovation and Technology Examples

Amazon.com makes for a good source of examples of the increasingly prominent role of technology even in business operating model innovation. In a *BusinessWeek* article, Jeff Bezos made the following points:

I think frugality drives innovation, just like other constraints do. One of the only ways to get out of a tight box is to invent your way out. When we were [first] trying to acquire customers, we didn't have money to spend on ad budgets. So we created the associates program, [which lets] any Web site link to us, and we give them a revenue share. We invented one-click shopping so we could make check-out faster. Those things didn't require big budgets. They required thoughtfulness and focus on the customer.⁷

And they require software. The associates program sounds like a business model innovation, but it is a business model that fundamentally is about tracking referrals and paying associates, and that's a whole layer of processing on top of online shopping. It is a model that pays directly for effectiveness in actual product sales, in contrast with most advertising spend, which only pays for consumer exposure. But it is a model that relies completely on software. Which came first, the chicken or the egg, the business model or the ability to create this capability using software? It's irrelevant. The only critical point is that the business seized the opportunity that technology created.

Now Amazon is pushing through the next wave of industry-shaping innovation, selling highly scalable, reliable, pay-for-what-you-use compute resources. Deciding that managing a highly scalable compute infrastructure is a distinctive competency for Amazon and that there's a big market for access to that competency was all about recognizing another opportunity to compete through technology. And here's the next Amazon marketing revolution that is totally keyed on its technical capability: Amazon has its tech star, CTO Werner Vogels, out talking up Amazon's key technical strategies and architecture at tech forums like *QCon* and *InfoQ*, as well as in his blog. Who better to sell to the technical community than highly credible tech

superstars, and who better to market compute cloud services to than architects and software developers watching the tech frontier?

One might say Amazon foreshadows what IBM characterizes as the enterprise of the future:

Hungry for change. Wildly imaginative. Disruptive by nature. Totally wired to the people who matter most. To some people, this might sound like your average teenager. In fact, these are the qualities companies will need to thrive in the near future, according to our newest CEO study.⁸

Social networking and its role in innovation and identity are explored in Clay Shirky's *Here Comes Everybody*.⁹ Recognizing that social networking affects opportunity discovery as well as product identity and customer mindshare, the IBM study reports that CEOs plan to increase investments 25% over the next three years to better understand socially minded customers, which chief executives identified as the fastest-growing trend.¹⁰

The enterprise is extended more and more these days to explicitly include customers who, through their interaction networks, stimulate product adoption epidemics and who help codesign, market, and support products through various social networking forums.

Who, then, should bear the charter to wonder, to find the opportunities to innovate, to deliver on CEO, and now also CIO and CTO, mandates to grow organically through innovation rather than by conglomeration?

FINDING OPPORTUNITY: INNOVATION SOURCES

Innovation comes from finding opportunities to add value in some new way. Whether we are talking about market-reshaping revolutionary/disruptive innovations or incremental and sustaining innovations, and whether we are talking about innovation in product or process, the primary sources are technology-driven, market-driven, and diversity-driven.

Technology-Driven Innovation

Technology-driven innovation comes from the development or integration of a new technical capability. This may be something fundamentally new (an invention) or an application of technology in a different domain. Productized, it is often a new capability in search of a market. Bucky Fuller¹¹ advanced the concept of technology gestation rates, pointing out that the timing of an innovation is important; it may precede market readiness and not get the traction its creators anticipated. Segways are an obvious example, for though they have found a foothold in airport and mall security,

they have not caught on universally, at least not yet. In a technology push style of innovation, opportunities to add value are sought through inventing or integrating new technology capabilities that excite the technologists. *Code Name Ginger* tells the story of the creation of the Segway, and the contagious excitement of creator Dean Kamen is captured vividly:

Engineers know that heat always flows from a hot object to a cold object. When Dean began talking about his passion for Ginger to a cool investor, you could watch the target's molecules heat up and start dancing.¹²

Market-Driven Innovation

Market-driven approaches, by contrast, seek to understand the customer and find ways to add something the customer will value. In *Hidden in Plain Sight*, Erich Joachimsthaler makes the point that the key to finding opportunities to innovate is understanding the “day in the life” of customers from the perspective of dreams, hopes, aspirations, passions, or sources of joy and delight, as well as concerns, fears, aversions, frustrations, or hassles — focusing on goals, not just actions and activities.¹³ Joachimsthaler uses this quote from the inside front cover of the J. Peterman catalog: “People want things that make their lives the way they wish they were.”

Enrolling customers in the innovation process is one of the central ideas in *Hidden in Plain Sight*. Although this insight originated in the consumer space, it translates just as well to business users.

In addition to looking at what people need or want, there's looking for the ingenious ways enthusiasts and problem solvers have addressed their own desires and needs. Consider the following example from Eric von Hippel:

A major auto company recently presented its “innovation road map” for the next ten years to a group of journalists and car enthusiasts. As the presentation progressed, it

became increasingly clear that some members of the audience were restless. Finally, one listener stood up and said, “Many of us have already built and installed every single one of the innovations you say you are planning to develop in the next ten years. Wake up and smell the coffee! Come out to the parking lot and take a look at what we have developed and installed in our cars!”¹⁴

This anecdote makes a key point that Patricia Seybold explores in *Outside Innovation*,¹⁵ and the whole notion of customer engagement is advanced in the cocreation wave. Again, this applies to IT as much as to product development. For example, if you're a business intelligence architect, look at what your company's leading competitive intelligence analysts are already doing, for they have been solving problems in an ad hoc way that may bear building into a solution that will increase awareness of, and responsiveness to, market opportunity.

Diversity-Driven Innovation

Both the market push and the market pull approach have worked. But if we want to make innovation more the way we do business, we need to look at putting insight into what is possible together with insight into what is desired, passion for what technology brings to the table together with passion for making a difference that customers care about (see Figure 2). Bringing together diverse perspectives, personalities, working styles, backgrounds, and proclivities is at the heart of the diversity-driven approach. *The Ten Faces of Innovation*, an outgrowth of the experience of leading designers at IDEO, is shaping thought in this area.¹⁶

We need to think in terms of multifunctional teams who are innovation and design centers. Instead, we typically have user interface folk who drive “user experience,” we have requirements or business analysts who drive “requirements capture” (as if they're running around and we just have to corral them), and we have software architects who wear beepers and are the second line of defense when the ops people can't put out

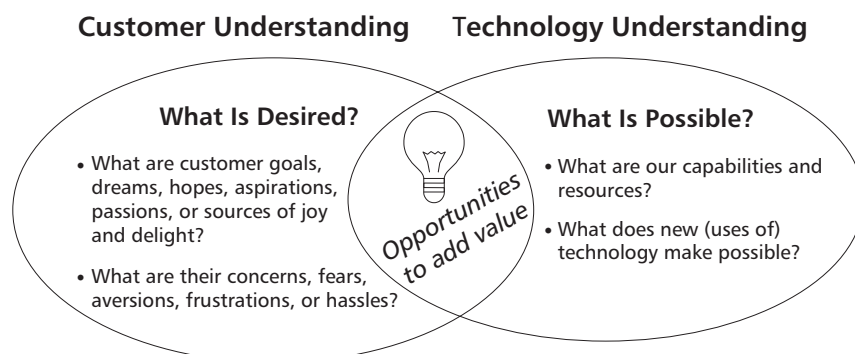


Figure 2 — Customer understanding and technology understanding.

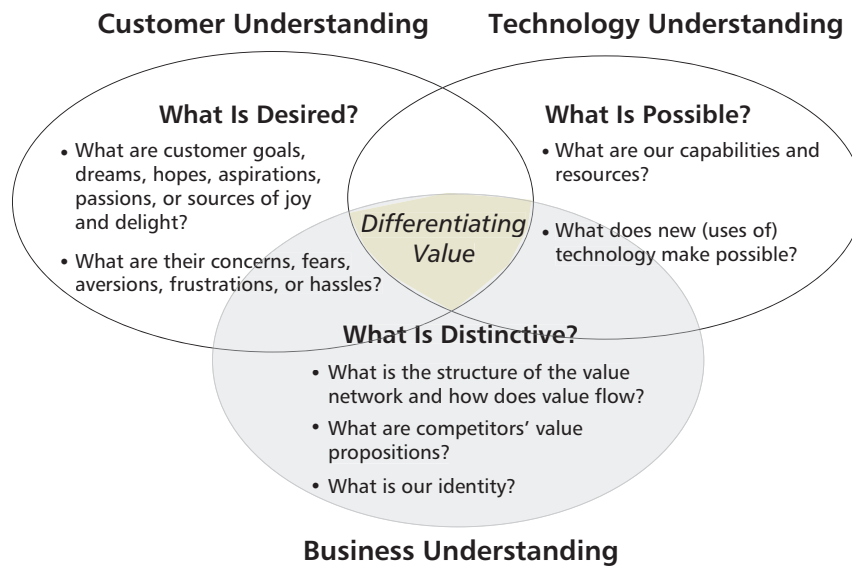


Figure 3 — Circles of innovation model.

on-the-line fires. Like these are all separable concerns. Different beasts — for beasts we make them.

Requirements aren't simply extant, waiting to be captured or even "discovered." They are invented, or at least, they should be — and are, when innovation happens. There is an interaction between needs/goals/ desires that exists along with ideas for meeting those needs, which changes the concept of the needs and the range of possible "solutions" to them.

Differentiation

We need to add another circle (see Figure 3) if we want to go beyond finding opportunities and head toward sifting for those that will be differentiating — those that will make our products or services, and our business, stand out in the minds and experience of our customers.

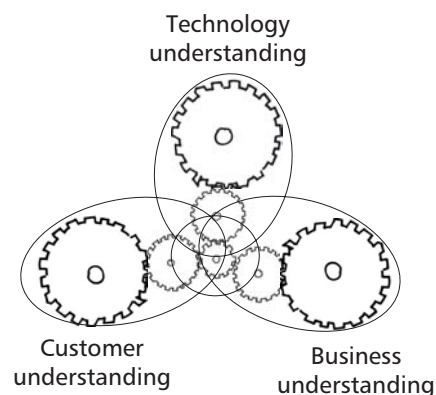
So who holds insight into customer/user needs? Who holds insight into our current capabilities and what new technologies make possible? And who has insight into our business positioning and direction relative to competitors and the value network? Yes, the center of expertise for each is different: marketing and business units, architects, and strategy planners and strategic management. Differentiating innovation happens when you put these together. Yet in most of our organizations, job specialization and role turf are pushing these circles apart.

What happens organically in the small, as in startups and incubators, becomes harder and harder to achieve in the large, where functions are separated and gatekeepers are placed, and place themselves, at the organizational divides that form between IT and "the business" or "marketing" and "development." The importance of

cross-functional teams comes up over and over again. Difference in perspective — diversity — can have its rough edges, but it fuels creativity.

We call this the *circles of innovation* model. Of course, like the circles that ripple out from a pebble thrown into a pond, each of these circles speaks to other circles — customers and those who understand them, technologists and those who have foresight into technology ripples well before others do, and value partners and competitors and those who understand forces shaping the marketplace.

So far we have addressed why innovation is on the executive agenda and why technologists, not just marketing and business owners, need to be involved in finding opportunities to create value. Architects, the bridge builders between the business and technology, and customers and technology-driven capability, are the obvious choice to be brought into, and even to lead, the core team that makes innovation a matter of practice, of ongoing corporate renewal. Next, we'll turn our



attention to finding and refining the opportunities that are the wellspring of innovation.

GETTING PAST “BUT”: HOW TO WONDER, WHERE TO LOOK

Quoting William Coyne, senior VP of R&D at 3M, Tony Manning writes:

Transformation is a group process ... and the leader who tries to control evolution cell by cell is not going to succeed. Never try to control or make safe the fumbling, panicky, glorious and chaotic adventure of discovery.

Occasionally I see articles that describe how to [rationalize] the invention process, how to take the fuzzy front end and give it a nice haircut. I would urge you to make that fuzzy front end as unkempt and as furry as you can because innovation is not neat. We stumble on many of our best discoveries.¹⁷

Which is not to say there is no hope of getting better at innovation, or that the process is necessarily so ad hoc that we should let it play out in a random, unguided way.

In our first Cutter report we began with a story, in the second we ended with one.¹⁸ This time we have a story near the middle. *The Wheel on the School*¹⁹ can be read as a children’s story, full of simple redemptive virtue, or as an allegory of the innovation process. *The Goal*²⁰ by Eliyahu Goldratt is a pivotal book in the Lean movement. We’ve been telling architects that *The Wheel on the School* is the hidden jewel of that genre — namely novelization of business fundamentals. Hidden, because it is a children’s story, and it takes a certain amount of brazen audacity to introduce a children’s story as an allegory for business leaders and innovators! It fits with the art of innovation theme of the July 2008 issue of the *Cutter IT Journal*²¹ — indeed it is art with an innovation theme! Besides, we believe it should be a pivotal book in the collaborative dynamic teaming movement. Is there such a movement? In software development, we see it instantiated as agile development.

The Wheel on the School is about vision and teaming and how core team members, with a shared vision and powerful vision stories, become the hubs around which flexible, fluid teams of teams form — like cogs, working together.

Pablo Picasso said, “All children are artists. The problem is how to remain an artist once he grows up.”

We might say children are open to wonder; the problem is how to wonder now that we are grown. How do we let go of our preconceptions, which confine and limit our thinking? Innovation is part awestruck seeking (to use a phrase coined by Grady Booch) and part

pragmatic engineering. *The Wheel* is about both. In what follows, we have to be necessarily more brief than the book deserves, but we hope to give you enough of a glimpse of the lessons that you will seek out this book that foreshadowed so much of the insight in today’s work on innovation and dynamic teaming.

How to Wonder

The first three chapters describe the activities leading up to, and the creation of, a shared vision.

Begin by Wondering Why

The story is set in a village in the Netherlands. Lina, the only girl in the school of six children, on her own initiative, writes an essay titled “Do you know about storks?” Her essay plants the seeds of a vision. Already in the first chapter, the diverse personalities and styles of the six children are evident, as is the gift of their teacher, a benevolent process coach and facilitator:

Eelka raised his hand. “But I’m afraid I can’t think much about storks when I don’t know much about storks. I’d be through in a minute.”

Everybody laughed, but the teacher’s eyes weren’t pleased. “True, true,” he said. “That’s right, Eelka. We can’t think much when we don’t know much. But we can wonder! From now until tomorrow morning when you come to school again, will you do that? Will you wonder why and wonder why? Will you wonder why storks don’t come to Shora to build their nests on the roofs, the way they do in the villages around? For sometimes when we wonder, we can make things begin to happen.

If you do that — then school is out right now!”

The team is chartered to wonder why, and let out of school an hour early. And so, powered by wondering, the wheel within the wheel of the book starts to turn, and things start to happen.

Finding Opportunities

The children all sat in a line outside and thought for a while, but that didn’t work and the boys went off to jump ditches. Lina, the lone little girl, looked into her wooden clog, and then, walking through the town with her “thinking shoe” in her hand, she noticed all the roofs in the town were steep. Grandmother Sibble III, the “oldest old lady” in the village, came out of her house to talk to the little girl, oddly wandering about with one shoe in her hand, stopping to stare at roofs. Lina explained she was wondering why storks didn’t nest in Shora, and mentioned the steep roofs. Grandmother Sibble said that could be remedied with a wagon wheel on the roof. When Lina said she’d thought of that, the old lady asked if she’d thought about trees. She hadn’t.

Would a stork think about trees? the old lady wanted to know. "It seems to me a stork would think about trees. And it seems to me that in order to figure out what a stork would want, we should try to think the way a stork would think."

Grandmother Sibble describes her grandmother's house, when storks had nested there long ago. She describes the full, systemic picture that had made that house attractive to storks: the roof, the wheel, the trees, a moat with fish in it.

"So you see you mustn't think our sharp roofs is the whole story, must you?" she said softly. "We must think about other things, too. Like our lack of trees, our storms, our salt spray. We must think about everything. And to think it right, we must try to think the way a stork would think!"

Grandmother Sibble tells Lina:

"There's a little tin on my kitchen shelf and in it there are wineballs. You get us each a wineball out of the tin. Then I'll sit on my stoop and you sit on yours, and we'll think about storks. But we'll think better each on his own stoop, because often thinking gets lost in talking. And maybe your teacher is right — that if we begin to think and wonder, somebody will begin to make things happen."

That candy tin has a picture of a village with storks on every roof.

Creating a Shared Vision

The next morning at school, the teacher asked "who wondered why?" and every hand shot up. Lina was furious because the boys all went ditch jumping, and she blurted this out, then felt bad for telling tales. The boys, and even the teacher, seemed angry. The teacher turned to Jella, the biggest of the boys, and asked him.

"Oh, I didn't think," Jella told the teacher honestly. "I asked my mother."

The teacher smiled. "Well, next to thinking, asking is the way to become wise. What did your mother say?"

"She says storks don't come to Shora because they never did. She says storks go back every year to the same nesting spots. So if they never came to Shora, they never will. So there's just nothing to be done about it, she said."

Lina excitedly recounted Grandmother Sibble's story about the storks that had nested in Shora when she was a girl, and described the house, the willow trees, the moat, the footbridge, and even the pickerel. At the mention of pickerel, Jella forgot he was angry with Lina, and all the boys had excited, shiny eyes. Then Eelka described how, yes, he'd gone ditch jumping but he'd fallen in a ditch and got soaked. He'd wished there were trees to dry his clothes on, then realizing there were no trees in Shora, concluded that was why there were no storks.

The teacher, facilitating, wanted to capture "The Reason Why Storks Do Not Come To Shora," and asked for the number one reason. Jella repeated his mother's assumption that storks didn't come because they never had. He was overridden, and they were about to conclude that the lack of trees was the primary reason, for that was what even Grandmother Sibble assumed.

Lina was frantic to break the cast of this assumption:

"Teacher," she said desperately to teacher's back, "even though Grandmother Sibble and everybody thinks it is trees, it has to be roofs. Storks don't just build nests in trees, they build their nests on roofs, too. But our roofs in Shora are too sharp! Oh it just has to be roofs," she pleaded. "Because we can put wheels on the roofs for storks to build their nests on, but we can't do anything soon about trees." Breathlessly she told the class about Grandmother Sibble's candy tin with the picture of the whole village on its lid and stork nests on every roof — because there was a wheel on every roof for the storks to build their nests on!

Pier and Dirk said almost together, "Oh, man, imagine a nest on every roof in Shora!"

"But that's just it. That's just it!" Lina all but shouted at them. "There's not a single wheel on any roof in Shora, because, like Grandmother Sibble, everybody must have figured it was no trees. So nobody ever put up a wheel. Nobody even tried! But how can we know if we don't try?"

The teacher asked if that could be, that if they put wheels on the sharp roofs, there'd be storks like there were in the picture.

"Aw, that was just a picture," Jella said scornfully. "You can put anything in a picture. All that is just a dream."

"Ah, yes, that's all it is," the teacher said. "As yet! But there's where things have to start — with a dream. Of course, if you just go on dreaming, then it stays a dream and becomes stale and dead. But first to dream and then to do — isn't that the way to make a dream come true? Now sit for a moment, picture it for a moment: our Shora with trees and storks."

Jella still resisting, pointed out that trees did not grow in Shora due to the salt spray, wind, and storms. They talked about protecting trees, but Dirk objected that would take too long.

"Making dreams become real often takes long," the teacher said. "I don't mean that it should be done at once. Our first problem is how to make just one pair of storks come to nest in Shora. That is what we are trying to do right now by first thinking out the reasons why storks don't nest in Shora. But after that.... If trees once grew where our school now stands, wouldn't they grow here again? Think of it. Trees all around our school!"

"And a moat with pickerel in it," Jella promptly added. "We boys could even dig it ourselves...."

"Yes, Jella, now you are getting into the spirit of it.... But first, before we can even start to think of all that, what must we do?"

"Find a wheel to put on the school," Lina promptly cried.

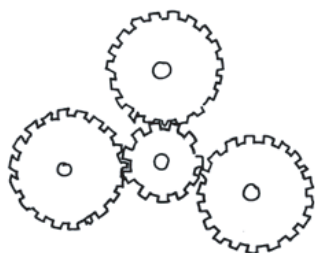
"Ah, hah," the teacher said. "Now we are getting to something we can do. Now do you see? We wondered why and we reasoned it out. Now we must do. Now we must find a wagon wheel, and then we must put it up on our roof. But behind doing that lies the long dream — storks on every roof in Shora. Trees! Maybe even a moat around the school. Can you picture Shora like that?"

Lina went and fetched Grandmother Sibble's candy tin with the picture of the village with storks nesting on every roof, and the teacher put it in front of the classroom, where all could see it.



Where to Look

The rest of the book focuses on finding a wheel and a pair of storks and installing these on the schoolhouse roof, or the first steps on the path toward the full vision. Having a vision doesn't mean the team enjoys an uneventful, deterministic path to success. With a clear mission — "find a wheel" — the team fans out, each child taking a different path away from the village,²² having been told to "look for a wagon wheel where one is and where one isn't; where one could be and where one could not possibly be." Even seemingly successful routes don't pan out as expected. The team has to regroup and start down new paths. Yet, with a vision to energize them, and to inspire others to join them in their quest, the children are ultimately successful.



The transformative power of great teams is beautifully conveyed in the story of Janus, a fisherman left legless by an accident at sea many years ago, who has become a mean recluse who children fear. He becomes a pivotal

team member, who helps the children recover a wheel and rescue a pair of storks at sea. The children help him find the best in himself, and he helps them find the best in them. The team is successful in fulfilling its vision. The village again has a pair of nesting storks, with their happy, lucky, clapping bills. Moreover, the children are richer for the experiences provided by good teaming and great mentoring, brought into their lives by the disabled fisherman, their teacher, and other villagers.

Debrief: What Do We Learn About Innovation?

This inspiring story reminds us of so many fundamentals. Innovation is a social process. People find the opportunities and create the vision, and many more people find the means to implement it. Our purpose is to create great products and systems that serve the competitiveness of our business by providing distinctive value to our customers. If our avenue to creating competitive products creates defining moments and personal growth for our team members, we are more likely to be successful. This book is full of breaking down the preconceptions we have about people and learning how working together, with all our diverse styles and differences in experience, creates fulfillment — personal fulfillment and enrichment as well as fulfillment of our aspiration to create value.

The Power of Wondering

The teacher was not thrown off by the pragmatic but dour response: *but* I don't know much; I'd be done (unsuccessful, but done) in a minute. He saw it as an opportunity to clarify what he wanted: not to think about what we do know, but to wonder about what we do not know. So much of our schooling, in education and in life, is about what we know and what is known. Yet to innovate, we have to get past that tendency to seek the safe ground, to focus on what we know. We need to be given encouragement and resources to wonder, to wonder why, to get beyond what we know, or think we know. But not for too long; timeboxing our wondering helps create momentum.

In the book *Code Name Ginger*, there's a story about Dean Kamen (inventor of medical devices as well as the aforementioned Segway)²³ that goes along these lines: Dean would ask "where would you put a third eye if you could have one?" and most people would say on the back of their head; without hesitation, generally, on the back of the head. But when Dean said "what about on the tip of your finger?" everyone agreed, without hesitation, that that was much better!

Henry Ford reportedly said, "If I had asked people what they wanted, they would have said faster horses."

In both of these cases, the transparent lesson is that customers aren't always the best authority on what the market needs. The deeper lesson, the lesson that *The Wheel* teaches, is that we need to be wary of our ingrained assumptions. We grow up with the "she has eyes on the back of her head" idiom, and that sets in place an unquestioned assumption about where to put a third eye. As soon as we relax the assumption, we open ourselves up to exploring other parts of the body as places to put a third eye. Assumptions can gate possibility. Our process needs to bring assumptions to the surface, into the shared consciousness of the team. Sometimes we need to explicitly push these assumptions aside and wonder what possibilities that opens up. And integral to *The Wheel* are lessons on diversity: a multifunctional team, with diverse perspectives and experience, helps expose assumptions and brings new solutions to the table.

Yes, when we start to wonder, it can be hard. Just thinking, wondering why, can be frustrating. How do we get started? Lina is very self-aware for such a little girl, and she knows that her clog is her thinking device. The old lady, too, knows that sucking on a wineball helps her think. These little rituals help get past that blank slate and start to fill our mind with ideas. For me, it is changing context, going to a coffee shop with blank tables (no phone, no e-mail, and no to-do list) with my lab journal with blank pages (no preconceived ideas).

The old lady and the girl talk and further each other's thinking, but then the old lady says they should each go to their own porch to think more, giving themselves more mental space in the quiet of their thoughts. This recognition of the importance of group work and individual work is exquisite. In software, to be sure we recognize the importance of solitary time with our computer, but it is often in uncluttered time that our hard work is brought to a moment of epiphany, that moment when a concept or solution jumps into compelling, heart-pounding, exciting focus. On the other hand, we allow ourselves to think that meetings are a time sink where no "real" work gets done. Yet important progress is made through collaboratively furthering each other's thinking, breaking past blind spots we have in our own minds. The old lady is important in this, for she is a kindred spirit who is excited about storks. She is a friend of the vision. If she had been neutral, or worse, negative, she would not have played the same powerful role in feeding the vision picture.

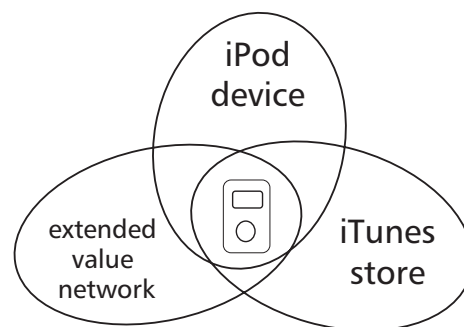
The book doesn't tell us "the way to wonder is to go and talk to your customers, and if you can't do that, find someone who has some experience with this set of concerns and talk to them. Moreover, if you don't

have access to a customer, you can still think like a customer." At least, it doesn't make these points in just those words. But it does make a powerful case for doing these things.

Think like a stork. That is the most enabling piece of advice, for it reminds us that we *can* take a stork's perspective. Even if we have an unprecedented system or a limited budget, we can empathize with our (figurative) stork, or if we just can't bring ourselves to do that, we can find the oldest old lady in our village to do that for us!

But there's more to it. *The Wheel* makes a compelling case for not just putting ourselves in the position of the customer, but also for thinking from the point of view of the customer, the user, and the ops team about the whole systemic picture, the "demand ecosystem" with all its interdependencies. It encourages us to think around the edges of our first hit on a solution, to think about what else.

When creating the iPod, it wasn't enough to think about playing music on a device more sexy than other portable music players. Making it more convenient than lugging CDs meant a whole new world had to be created around licensing, downloading, selling and managing digital rights, and a truckload of other stuff I haven't hallucinated. The whole experience had to be thought through, to be created, first in the minds of men and women, for the solution didn't exist yet. Analogy pieces to it did, but the larger ecosystem had to be envisaged to be built. The ideas had to be iteratively explored to understand what it would take to make the new ecosystem not just viable but also compelling and to create an experience that delights.



Creating a Shared Vision

Suggesting it would be nice to have storks nesting in Shora isn't yet a vision. The vision is not simply the motivating opportunity but a *shared* picture of the desired state. The opportunity is integral, but the vision — storks nesting on wheels on all the high-pitched roofs in the village, with trees, and the schoolhouse also with

a wagon wheel and a pair of storks, trees, and a moat with fish in it — fleshes out other dimensions of the desired state. The vision illuminates key elements of the solution: a pair of storks, a wheel, trees, water, and fish. But it doesn't address how these come to be there. It paints a picture that speaks to the hearts and minds of the core team — the children — and later the extended teams of teams — the villagers and others who come to play a role. And it indicates key steps to be taken, without dictating how to accomplish them.

What is so important about fish? They are important to the storks, who feed in the marshes. But Jella lights up and becomes motivated at the thought that this vision has fish in it. It isn't just about storks but fish too. Remember Jella saying, "but I don't know much"; "but that's just a picture, just a dream." Having something he cares about enter the vision picture motivates him to embrace the whole picture, and he engages wholeheartedly in the action that follows.

We have become so "do" focused that we forget "first to dream, then to do." Nike's "Just do it" and IBM's "Stop talking. Start doing" commercials, countless blogs, and many in the agile movement all emphasize doing. This book points to the essential balance allowing just enough envisioning to direct the action toward a concerted purpose that the team embraces. As new team members are initiated into the group, vision stories are told to bring them quickly into alignment with the common purpose, energized by a vision that they find compelling too. The vision paints the "big picture," the ecosystem we're trying to build or adapt, and illuminates key incremental stages to building out the vision — doable chunks, user stories if you like, that will fit into the vision picture.

Where to Look

One of the key innovation principles in the book is captured in the teacher's instruction: look where you would expect to find a wheel and look where you are sure you could not possibly find a wheel. This principle serves as much as a guide to innovative solution-seeking as it does to seeking innovative value-creation opportunities. We need to open ourselves up to possibility. Sometimes possibility comes from working closely with end users and delivering code that fits the stories they tell. And sometimes we just have to get out and find new stories, shape the world before it boxes us in with preconceived expectations. Look in unexpected places — impossible places as well as expected places.

A number of the exploits around finding a wagon wheel end in failure and serve to remind us of the importance of failure. So important is failure to innovation that Randy Pausch created a "first penguin" award²⁴ to help his students grok that while success is better, failure is next to success and failing fast, to learn fast, should be embraced and even rewarded.

Changes in the environment create a sense of urgency for the team; storks are starting to fly overhead, first in pairs and then soon in greater numbers, as many as 14 at once. There is a window of opportunity, and if the team misses it, its dream is lost at least for this year. But the team doesn't lose focus and works on fulfilling the first element of the vision. The vision is broken up into steps that can be tackled one at a time. And each step, though difficult and challenging, is not impossible, and, step by step, through hard work, great dynamic, self-organizing teaming, and some luck, they ultimately succeed in putting the first stage of the vision in place. The first release is the school with its wagon wheel and a nesting pair of storks.

One could read *The Wheel* as just a lilting, classic children's story; a story about climate change and community action to restore the environment; or a story about the more general process of innovation.²⁵ Jonathan Swift's *Gulliver's Travels* was reportedly read from the nursery to the British Cabinet. *The Wheel* is not a political satire, but with the climate crisis on national agendas, and innovation on corporate agendas, it arguably should be read from elementary school to R&D labs and IT shops to government. It reminds us that, with child-like enthusiasm for doing a good, right thing, we can set the wheels of change in motion to accomplish the big things we dream of. First, we have to dream. Then, we have to act. Break the problem down into doable chunks and act. It is a simply conveyed yet extraordinarily sophisticated exploration of humanity and the collaborative problem-finding/problem-solving process that is the hub of innovation. It is a story about the people we have in innovation processes, in system or product inception, and in development: dreamers and optimists, pragmatic problem solvers and doers, even curmudgeons, who are all transformed, aligned, and impassioned by a vision they embrace.

In the second half of this report, we draw on the lessons of the first half but focus on specific things to do as we develop the system concept and surface and address dominant uncertainties, challenges, and risks so that we can orchestrate the socio-technical process of building our innovation vision effectively.

DIFFERENTIATION BY DESIGN: FINDING OPPORTUNITY

Strategy identifies differentiating value propositions to shareholders, customers, employees, and the extended value network.²⁶ Finding distinctive opportunities to add value, the cornerstone of the product and business strategy process, needs to be more a matter of intention and less a matter of happenstance. From *The Wheel*, we have learned key principles of an innovation process, and in the sections that follow, we describe a process that applies these principles together with “Pareto tools”²⁷; sifting through our experience, these are the tools or techniques that we have found hugely productive in team settings yielding high insight-to-time value and energized, innovative teams.

If you are familiar with the Visual Architecting Process (VAP)²⁸ you will recognize these as the early phases and cycles of VAP. In the coming sections, we explore opportunities to create value and create a compelling, shared vision for the system (the Initiate and Gain Commitment phase of VAP; see Figure 4).

Finding the Opportunities to Create Value, or How to Wonder

In preparation for creating the vision, we need to bring together people with diverse perspectives to explore, and we need to encourage divergent “out-of-the-box” thinking.

There’s an old Brazilian proverb that when you dream alone it’s just a dream, but when you dream together it’s already the beginning of a new reality.

But the power of “dreaming together” can be hard to harness. The graphical facilitation tools that we find extraordinarily helpful here leverage the Group Graphics® work of Grove Consultants International, and we strongly advocate their work in vision and strategy formulation, team development, and facilitation.

Gathering Input

As we start out gathering input, it is useful to hold a launch event with core members of the multidisciplinary product or system concept team and other key stakeholders, chosen for their diversity of perspective, experience, and style. Competitive Landscape Maps help bring these perspectives and insights into a shared picture and help expose assumptions that would otherwise gate possibility.

Competitive Landscape Maps

In our 2005 *Executive Report*²⁹ we described our Visual Strategy Framework. This framework, along with ideas from *Hidden in Plain Sight*,³⁰ Grove Consultants’ Context Map Graphic Guide®, and our own experience, informed the creation of our Competitive Landscape Maps. These pull together onto one great landscape view, elements that help illuminate and motivate opportunities to create differentiating value, including:

- **Our market segments** — customers by use context and their goals, concerns, aspirations, frustrations, activities, and so on
- **Our strategy** — identity, value propositions, and capabilities
- **Our competition’s strategy** — identity, value propositions, and capabilities
- **The winds of change** — market and technology trends
- **Environmental forces and factors** — legal requirements, economic forces, and so on
- **Uncertainties and threats**
- **Challenges we face**
- **Opportunities**

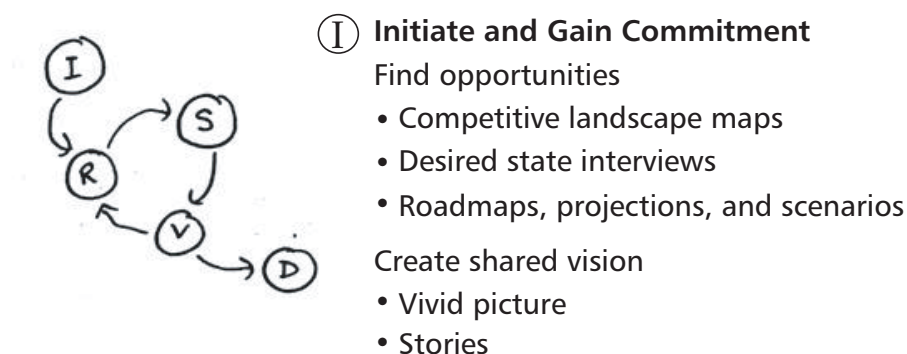


Figure 4 — The Initiate and Gain Commitment phase of the Visual Architecting Process.

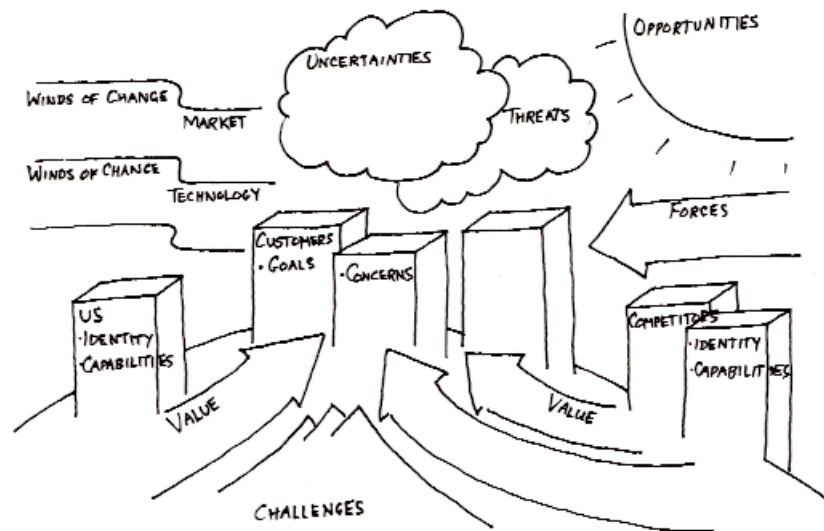


Figure 5 — The Competitive Landscape Map template.

The Competitive Landscape Map gives a big-picture view of the business context, including the demand ecosystem (see Figure 5). It provides for structured brainstorming and, when the participants are selected for diversity of insight and experience, can really stimulate the creative foment that produces new ideas.

All of this information could be captured in tables. But the graphic template pulls information together into a *shared* big picture, bringing different perspectives into an integrated view. We need to recognize that we are creating socio-technical systems; both the process and the product have social and technical aspects to them. It is foolhardy to think that the technical complexities are the only concerns. Subtle cues, like “this is our world view,” “our map,” or “we’re all on the same page,” lend themselves to fostering collaboration. Visualizations like this, and later stakeholder profiles, use case diagrams, and conceptual architecture, as well as other architecture models, create the shared thought space that helps the team surface assumptions and stimulate creativity. They help us think individually — as Colin Ware points

out: “Although we can, to some extent, form mental images in our heads, we do much better when those images are out in the world, on paper or a computer screen.”³¹ Moreover, they help us think as a team.

Stakeholder Input

The system concept team needs to get out and talk with stakeholders, including customers (clients, users, etc.), seeking to understand their tangible goals and intangible dreams, sources of joy and delight, and business issues, concerns, frustrations, or hassles — not just their day-to-day actions and activities. And we need to look to our innovative customers to find what they’re already doing to “make their lives the way they wish they were.”

Ideas will come from a variety of sources: the strident voice of the dissatisfied customer, the eager voice of the customer on the bleeding edge, the fading voice of history and alluring whisper of the future, and the ingenuity of the design and development team.

The architect, as leader of the design and development team, needs to hear these voices, in person at least some of the time, to be inspired and inspiring. They are the source of stories that will shape the vision of what is possible and speak to and win the hearts and minds of sponsors and developers, marketing and project managers, and all the others that play a role in bringing innovation in process and product to realization and eager adoption.

Desired State Interviews

We have found desired state interviews to be a fruitful way to gather stakeholder input to the vision. In



essence, we ask key stakeholders to “move into the future when the system is delivered” and think about how things “are” now that the system is in place and delivering value. Alternatively, we can take Russell Ackoff’s³² approach and ask people how they would like things to be today. In either case, we need to state two important rules of the game:

- 1. Must be feasible and viable.** We need to restrict ourselves to what is technically feasible and operationally viable, so we remove Utopian dreams from scope — no telepathy or “beam me up Scotty” transporters, no flying cars.
- 2. See past the obstacles.** We need to free ourselves from the obstacles of the present. We tend to want to solve the problems we currently face, and that distracts us from making strategic moves that would change our positioning.

What is important is to envision the system (organization, product, whatever we’re setting out to build or improve upon). We want to be inventive, though not unrealistic. And we need to think of the system as a piece of bigger systems. It will impact and be impacted by other systems, and we need to think through and do what we can to foresee implications and hidden assumptions that will otherwise surely surface to bite us.

A desired state interview frees the person being interviewed from all the clutter of today’s problems and lets the individual think about how he or she wants things to be. Then we look at how to get there, from here (see Figure 6); at what obstacles were overcome, and what changes were put in place.³³

We have to do some work to think around the edges of what we know, find out new things, look at trends and

patterns, look at other geographies or industries, look at our markets with a different lens, and stretch what we know. You know the adage, “If we keep doing what we’ve always done, we’ll get what we always got?” To that we need to add, “If we act on what we already know, we’ll keep doing what we’ve always done, and get what we always got.” So we need to find out new things. Call it going to the future, call it staying in the present, or call it whatever. Just get out of the box!

Roadmaps and Scenarios

Architecture or technology roadmaps are a key input to the strategic visioning process, for they illuminate opportunity and risk. That said:

Prediction is very difficult, especially about the future.

— Niels Bohr, physicist

Nonetheless:

Programming is the act of deciding now what will happen in the future.

— James Coplien and Neil Harrison³⁴

Given that we create today what will be used in the future, it behooves us to think a bit about what that means: not to be paralyzed by uncertainty, but to assess likelihood and risk. To do so, we need to build anticipatory awareness, paying attention to trends and events in our market and technology space.

We map out our product or system release plan, our own internal development plans for infrastructure (platform) development, our plans for capability development, what we know of our vendors’ release plans for technologies we depend on, and so forth. We might call this our roadmap, though strictly speaking even this is

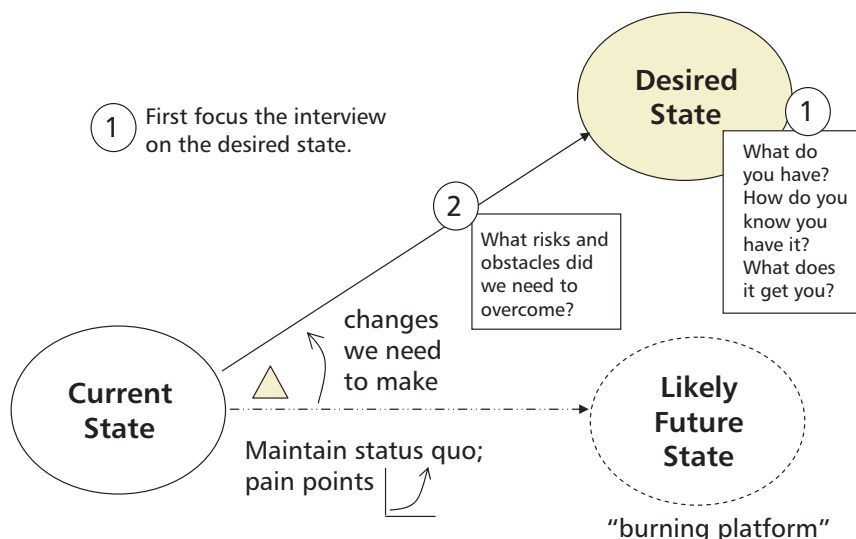
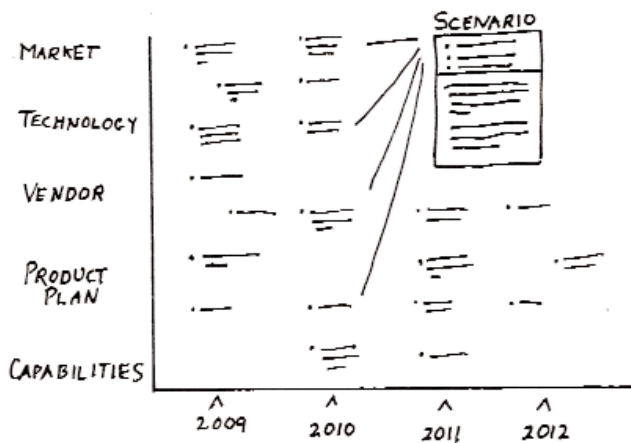


Figure 6 — Change map.

projection. Still, these are intentional, and though obviously not deterministic, we will invest in making this future unfold, so it is a more likely future than most other futures we could speculate about.



But then we also want to overlay these plans with projections about how current trends might unfold. So we have moved beyond “product roadmap” turf into more reasoned projections and educated guesses about the future of markets we compete in and the technologies that might well (re)shape these markets.

Why? Because our architecture has to see us through that world of tomorrow — when the system will be fielded and evolved. Today’s world may well be a better predictor of that world than any other prediction we might come up with. But we still need to think about what the future is likely to bring. Even if we had a perfect crystal ball, we couldn’t build all of the functionality, nor all the structural support, we will come to need into our first release — if we want to release in competitive time frames, anyway. And we don’t have a crystal ball, so we can’t build to meet every contingency. But it also doesn’t make any sense to act like an ostrich, bury our head in the sand, and ignore likely threats and opportunities. Consider the following pieces of advice:

Predicting the future may be impossible, but ignoring it is irresponsible.

— Eberhardt Rechtin³⁵

I often describe the life of a software architect as a long and rapid succession of suboptimal design decisions taken partly in the dark.

— Philippe Kruchten³⁶

Max Bazerman and Michael Watkins³⁷ point out that organizations are vulnerable to surprises in part because of our human predilection for harboring illusions that things are better than they are, giving weight to evidence that supports our preconceptions, giving

more weight to maintaining the status quo, and preferring to avoid some pain today than prevent a lot of pain tomorrow. Information and responsibility in organizations is fragmented across organizational silos, and this too makes organizations vulnerable to predictable surprises. To counter this propensity to be blindsided by disasters the organization should have seen coming, Bazerman and Watkins encourage scenario planning and risk assessment:

Scenario Planning: Assemble a group of creative knowledgeable people from inside and outside the organization. Review company strategies, and identify external trends, critical business drivers and potential flash points. Based on this analysis, construct a plausible set of scenarios for potential surprises that could emerge over the next few years. Include scenarios that, though unlikely, could have a big impact.

Risk assessment: Identify risks, assess probabilities of these events, and estimate the costs and benefits of particular outcomes.³⁸

To help find opportunity and risk, we recommend looking for scenarios where the turn of events is really, really good for us (e.g., demand takes off at an unprecedented rate) or really, really bad for us (e.g., the recession is deeper and longer than anyone expects).

The flip side of value is risk and challenge. If we’re seeking to innovate, we’re taking on some degree of risk. And if we don’t innovate, we’re taking on risk. The key, really, is to become better at identifying the risks that pose the greatest and most likely threats and to manage them (to avoid, transfer, reduce, or mitigate the risk). Risk aversion is, in part, due to lessons taught in the school of hard knocks, and, in part, it is intolerance for the ambiguity we face dealing with fuzzy futures. But the architect, as leader into the future, has to do some scouting out of that future terrain he or she is leading the team into, and we’ve found that roadmaps, projections, and scenarios help the architect take a long view. But this doesn’t mean to attempt to take care of every detail of tomorrow today, for that is so futile as to be a sign of insanity! We carry a spare tire in the car, but not a spare battery. It is a matter of finding the balance for our project. Spending some time looking at how market and technology trends could play out, and weaving a set of scenarios, helps alert us to opportunity and risk.

We need to “go into the future” and “think around the edges” just enough to think through system deployment scenarios so we don’t build production capacity for 40,000 Segways per month when annual sales were something like half that over the first five years, or the converse with Twitter, the social networking service, which, earlier in 2008, faced issues with its architecture

not supporting dramatic scalability demands. We can't predict the future, but we can make reasoned arguments about alternative futures³⁹ to assess the opportunities and the threats that those alternatives expose us to.

There is significant complexity even in our determined past and so much more so in our undetermined future. We need to strive to simplify; however, to reduce life to binaries (e.g., you either talk or you act; talking about tomorrow prevents us from acting today; act today and trust that tomorrow will care of itself) is to sweep aside all the complexity that is the rich fertile mess on which innovation thrives. We are not trying to conquer complexity by hacking off its limbs (a Monty Python skit comes to mind) but rather to harness it. As Howard Behar says: "Think like a person of action, and act like a person of thought."⁴⁰

Create a Shared Picture of the Desired State

Through the process of exploring the Competitive Landscape, we identify value-creation opportunities, we explore their position relative to competitors' value propositions and identity, and we begin to get a sense of the challenges and capabilities we would need to build. Through desired state interviews, and possibly other techniques such as customer shadowing, we get a richer sense of the opportunity. Of course, we don't take the "why" in "wonder why" literally! Yes, we wonder why: why is the customer not delighted, why does the customer have aspirations and hopes that are not fully satisfied in this area? And we also wonder what: what would make the customer's life more the way he or she wants it to be? And "what if?" and "what else?"

As an innovation comes into focus, we don't stop there. We think about the systemic picture into which it fits and think about how it impacts and is impacted by that ecosystem, in part so that we can think through other innovations that we or value partners will need to put in place to create the environment in which our innovation will thrive. And because innovations don't exist in the past, nor even the present since they take time to birth, we explore the future using our roadmap and projections, as well as scenarios, just enough to get a sense of the critical pressures and opportunities that will be made or broken for our system when it is deployed and over the first generations of evolutionary pressure to which it will be subjected.

Once we have done a pass at this exploration, our ideation or system concept team will be full of ideas, but these ideas need to be sifted, evaluated, prioritized, and synthesized. It is not enough to "wonder why," to gather ideas for creating compelling, distinctive value. A cohesive shared vision must be created. Consider the

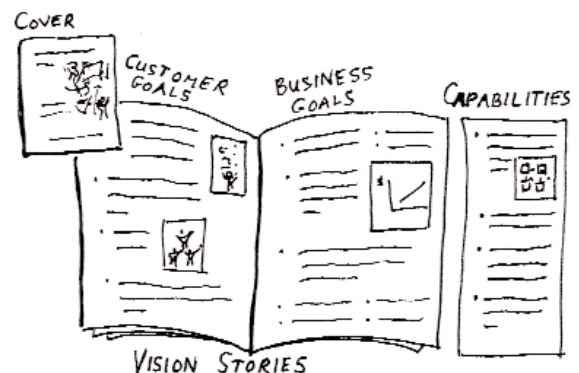
following quote by French author Antoine de Saint Exupery: "If you want to build a ship, don't drum up the men to gather wood, divide the work, and give orders. Instead, teach them to yearn for the vast and endless sea."

This quote resonates, but not because we think that sailors necessarily make better shipbuilders. It is, instead, that the idea of creating a shared yearning, a common quest, is so important to communal creative endeavor. Then, building the vessel that will take us where we want to go will be an inspired venture that generates goodwill enough to surmount the challenges. This is not about getting into a cozy circle around a campfire and singing "Kumbaya." If we are creating a complex system that requires expertise that is distributed in different people's heads, we need to create a shared understanding of what the system is. But this is more than just about shared understanding. This is about enrollment, drawing in the creative contribution of someone as "left brain" and pragmatic as Jella and enabling someone as little and unlikely as Lina to become a leader of boys and men, not by commanding but by inspiring.

In *The Wheel*, the vision was both a vivid picture and a set of compelling stories, integrating elements that various key stakeholders, including team members, were personally excited about.

Painting the Vision Picture

As we create the vision picture, we want to be thinking about the system within its context or, for many systems, within its various contexts, of use. And we want to be thinking about the system from the point of view of the business, including stakeholders in business success such as shareholders and executive sponsors, as well as the people who will develop, deploy, maintain, and operate the system. Moreover, we want to capture vivid stories about value, stories about how customers' lives will be better, how the system will delight users, how it will ignite customer enthusiasm, and how it will create competitive advantage.



The technique we have found most valuable, is what Grove Consultants (and others) call a “Cover Story Vision.” It is more than the press release we envision sending out at product completion. It is the story that would be published in an influential newspaper, journal, or magazine in our domain. During our visioning session, we gather images and headlines for the cover and central stories and quotes from excited customers, our CEO, and our ops team. These will be direct quotes and images from the desired state interviews and customer visits, as well as new ideas that come out of the visioning session, from the generative foment that happens when a group of people with diverse perspectives and experiences collaborate to advance the group thinking that is being vividly pictured (literally) across the breadth of a wall. In the sidebar, we draw out challenges and risks that “were” (in the future time frame of the cover story) overcome and outline key capabilities and properties in the architecture and the organization that helped us address these challenges.

Communicating the Vision

The vision work is not done when the team has a sense that it has a cohesive vision that the members are aligned around.⁴¹ As more and more people are brought into the development process, the vision is used to help them get their heads and hearts into the game, to help them understand the destination and what it calls for, and also to be motivated to pour their creativity into all the contributing innovations that will be needed to realize the overarching innovation vision. So there is important work in communicating the vision in a way that draws people to invest themselves in refining, elaborating, and building it.

In *The Wheel*, the children used both the vision picture and stories about storks coming from their African habitat to nest in Dutch villages, bringing luck. Stephen Denning has done a masterful job of positioning storytelling as a powerful tool to engage and lead.⁴² Yvon Chouinard also notes the importance of rhetoric in *Let My People Go Surfing: The Education of a Reluctant Businessman*:

A chief in an American Indian tribe was not elected because he was the richest or had a strong political machine; he was chosen chief because of his oratory skills, which were invaluable for building consensus within the tribe.⁴³

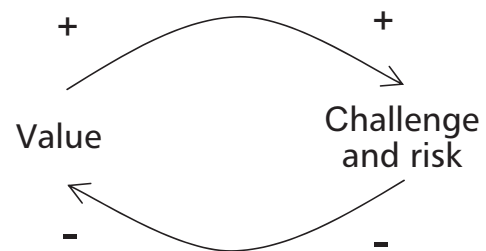
In addition to telling crafted stories, Google’s Tim Armstrong advocates drawing the vision while you talk, pointing out that using a simple sketch drawn in real time helps communicate more effectively than a slew of slides. While Armstrong is in marketing (he’s the

president of advertising and commerce, North America, and VP, Google, Inc.), we shouldn’t dismiss this as just a marketing thing. It’s a communication thing. It is worth remembering these heuristics from Reichtin’s field-launching classic *Systems Architecting*: “Never stop talking about the system”; and “There’s no such thing as immaculate communication!”⁴⁴

Dan Roam has been promoting using hand-rendered sketches in business communication.⁴⁵ He points to a feature of Microsoft PowerPoint that allows one to draw live on slides in presentation mode, so we can project dynamically drawn sketches and models even when we are addressing larger audiences. When you draw the vision (and later the architecture models) live, it unfolds as you talk, allowing you to manage attention, but also to work with the input you get, increasing participation and understanding. As Confucius purportedly said: “Tell me, and I forget. Show me, and I may remember. Involve me, and I will understand.”

DIFFERENTIATION BY DESIGN: MAKING IT HAPPEN

The vision tells us what we’re looking for, but not where to look. Strategy sets the destination and the vision and also identifies the direction or high-level path to attaining the vision. So our strategy identifies the capabilities that must be built to deliver the value propositions that are central to the vision. Since value comes hand in hand with challenge and risk, our strategy should outline approaches for dealing with the key challenges and risk. But further, our strategy has to balance the value we strive to achieve with the challenge and risk we take on, and scope is an essential outcome of strategy setting.



In a simple situation, the direction may be self-evident from the vision: find a wagon wheel, install it on the schoolhouse roof, find a pair of storks, and install them. That’s not to say that finding a (figurative) wheel is trivial, nor that finding a pair of storks is a foregone conclusion. It’s only that the first step is already evident, and it is clearly partitioned from the step of finding the storks.

As more complexity is added, the strategic alternatives expand. The innovation process proceeds through a series of divergence-convergence cycles (see Figure 7). We already saw the first cycle, with divergent, out-of-the-box thinking being encouraged and facilitated through Competitive Landscape Mapping, desired state interviews, and roadmaps, projections, and scenarios and then being brought to convergence through the creation of a shared vision.

Further divergence-convergence cycles are needed to discover and explore alternatives then set direction through scoping the system and deciding what capabilities to build to deliver key value propositions that will differentiate our product, service, or business in the market.

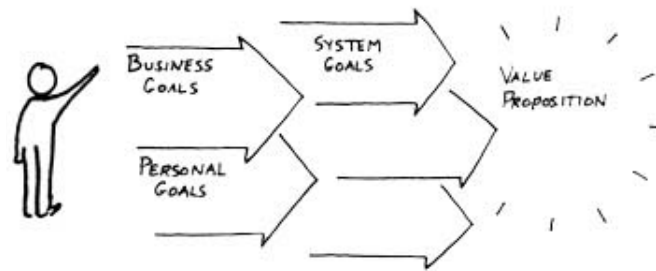
Where to Look: Scope and Direction Setting

The next major cycle involves doing enough value exploration and discovery to set the product and technical strategy, deciding on system scope to focus subsequent divergence-convergence cycles.

Applying key principles that underlie agile development, namely rapid iteration and stakeholder participation, we cycle quickly through exploring and better understanding system value propositions (which will become formalized⁴⁶ as functional and nonfunctional requirements) and what it will take to build them.

Elaborating Value

Again, we like Pareto tools and find that stakeholder profiles fit that characterization. Stakeholder profiles direct



attention at, and capture, the business, personal, and system goals of specific stakeholders and representative stakeholders in different use and business contexts.

Use case diagrams can be used to create a big-picture visualization of the goals the system will support, with more detailed views as needed to explore avenues of opportunity. They can be used both as a structured brainstorming device to gather ideas for goals to support and, later, as a scoping tool to decide which goals to out-frame and which goals to support in what time frames through staged delivery of capabilities. Of course, use case diagrams are not the only tool in our tool belt, and we'll use story boards, user stories, process diagrams, and other vehicles as we see fit given our situation and the dominant uncertainties we are addressing.

The Circle of Excellence

As we explore user experience⁴⁷ design we are finding the dimensions along which we should excel to surprise and delight customers. We cannot surpass the market on every possible dimension, so we have to figure out what is exciting to our (would-be) stakeholders.

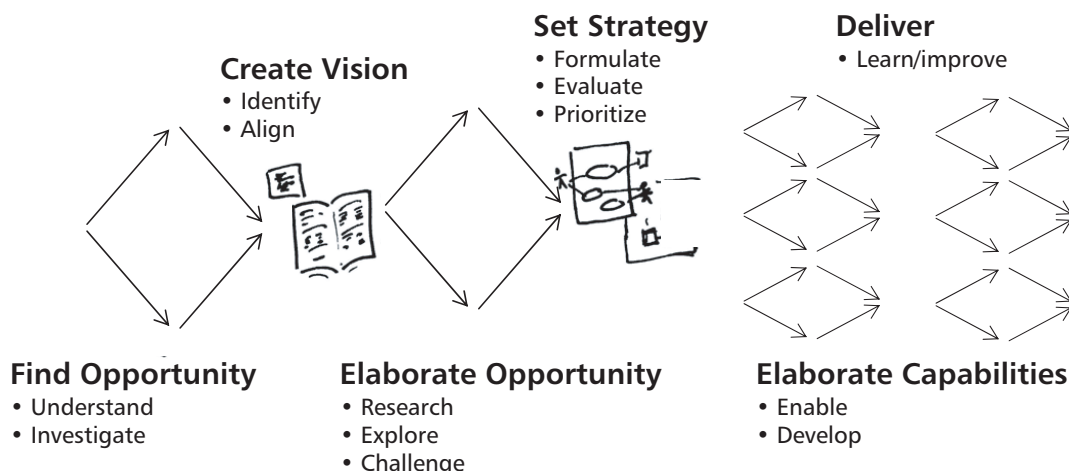
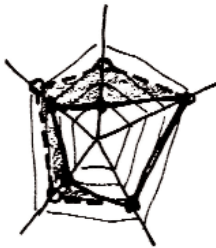


Figure 7 — Divergence-convergence cycles in the innovation process.

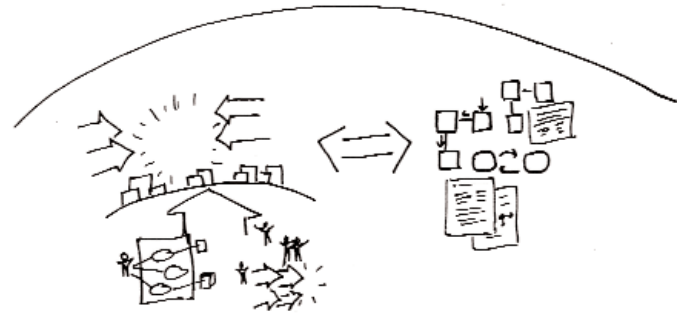


A “circle of excellence” (visualized as a Kiviati/radar/spider diagram) brings to the foreground the fact that for customers and for our business stakeholders, including development engineers, excellence is compelling, and we need to pick the dimensions along which we will strive to excel — that is, what capabilities are within the circle and what are the properties that define excellence for our product. This is very much about scoping and being clear where good enough is just that. This is not just a consideration for high-end products with design sizzle, but also for a low-end product where, for example, ease of use, core features, and robustness may be within our circle of excellence — where beating customers’ expectations has to do with lowering the usability barrier, not just the price.



Design Across User Experience and Structure

Scoping decisions⁴⁸ need to be informed both by opportunities to add value and by challenges inherent in doing so; adding capabilities to enhance the value proposition adds complexity, challenge, and risk. To get a system built in competitive time frames this duality has to be understood not just to sanity-check system feasibility but also so that system scoping and staging decisions can be made realistically. Moreover, the process of exploring the architecture, or system structure, illuminates opportunities to add value. While we are quickly and cheaply iterating through system concept development using sketches, modeling out loud in small groups, and with varied stakeholder participation, we can cover a broad space of options both in terms of value or utility and in terms of structural designs. As architect Frank Lloyd Wright said: “Form follows function — that has been misunderstood. Form and function should be one, joined in a spiritual union.”

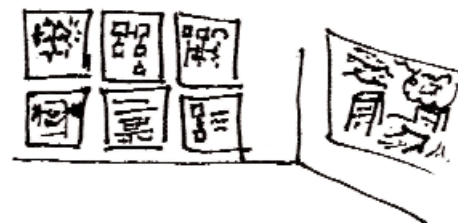


Therefore, we strongly advocate iterating between exploring system concept/high-level user experience design and system structure design. This means iterations through architecture strategy and conceptual architecture, certainly, but possibly also dives into logical and execution/physical architecture⁴⁹ as needed to understand some area of challenge or risk to chart a strategy for dealing with it. Moreover, dives into code may be necessary to better understand some key aspect of the user experience/value proposition or to explore and demonstrate an approach to addressing an area of architectural challenge or resolving a risk. However, the discipline is in staying lightweight, where possible using models to support thinking and making reasoned arguments.

Just Enough Design Up Front

Through these iterative cycles, we direct our attention to exploring and resolving dominant uncertainties, challenges, and risks. As Buckminster Fuller said: “I always say to myself: What is the most important thing we can think about at this extraordinary moment?”

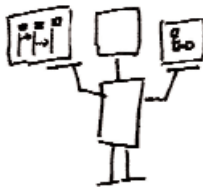
The central idea could be called “*just enough* design up front” (JEDUF). How one would characterize “just enough” depends on the complexities⁵⁰ of the system (degree of novelty, intrinsic difficulty of the subject matter, scale and/or scope of the system, system dependencies, etc.) and the development environment (large numbers of developers, geographically distributed teams, disparate organizational cultures, vendor dependencies, the need to integrate specialized expertise in various domains, etc., all drive complexity up). Suffice it to say, if we embrace higher degrees of complexity in order to deliver more aggressive value propositions, we need to do more design up front so we are better positioned to manage that complexity.



JEDUF for a system that will be designed by a small team in a short time frame (on the order of a few months) may be a day-long session where the team covers the walls, sketching stakeholder profiles, use case diagrams, conceptual architecture models,⁵¹ and key architectural mechanism designs. JEDUF for a highly novel system that will disrupt the marketplace, or for a complex system that will take dozens or even hundreds of people to build, may extend to months, for a key idea here is to fail fast and cheap, by exploring down various paths enough to get a sense of where the high-payoff route lies.

Architecture as System Design

Architecture is about design of the system, not just design of the structure that allows the design of the system to stand up to stresses and strains of the load placed on it. Architecture impacts the design of the user experience, not in the small like interior decorating, but in the large like layout and flow, supporting function with qualities that delight, where it matters, and satisfy, where that is good enough. Architecture constrains and enables behavior, and architecture impacts properties of behavior.



As a group from MIT writes: “Architecture is important to both naturally occurring and overtly designed systems since architecture conveys behavior.”⁵²

Architecture decisions are those we need to think most carefully about, because they have strategic impact (they impact our value proposition and competitive position) or significant structural impact and bear a high cost of change.

Architecture decisions are design decisions with the greatest consequence if we get them wrong. These include early decisions that gate or constrain all subsequent decisions.

In particular, architecture decisions include:

- Architecture strategies (e.g., principles, pattern selection), designed to address architectural challenge and risk
- High-level system decomposition/composition decisions (e.g., architectural elements, their responsibilities and externally visible properties [and interfaces], and interrelationships)

- Architectural mechanism design, addressing architecturally significant cross-cutting concerns

These decisions impact scoping decisions (what value propositions to offer; what capabilities to build to deliver these value propositions) and staging decisions (what decisions to make early; what capabilities to build in what time frames).

Visual Modeling

Modeling is integral to architecting. It is a critical thinking tool to support decision making during complex system design.⁵³ Besides, your head very effectively encapsulates your thinking, unless you speak, draw, and write. So you at least need to speak and draw to create a shared mental space where others can influence and get access to your thinking. And now we have a chance to make it “our” thinking, our architecture, because it is shared, not just in one head.



Just as pair programming typically produces better quality and overall productivity, there is value to having a small team work on the architecture. Still, one wants to be fluid, sometimes working alone concurrently to produce different variants that can be assessed for relative merits, and sometimes working in a small team, using the visualizations as the shared thought space to surface assumptions and stimulate creativity.

Improvement and Validation

It is important to expose the evolving worldview, the context, vision, system concept, conceptual architecture, and so on, to early and frequent input from other stakeholders (lead customers, marketing, other architects, developers) who can add ideas and throw questions at the design.

Alfred Sloan, who ran GM from 1923 to 1956, was onto something when he said at a meeting, “Gentlemen, I take it that we are all in complete agreement on the decision here. Then I propose that we postpone further discussion ... to give ourselves time to develop disagreement and perhaps gain some understanding of what the decision is all about.”⁵⁴

Temple Grandin tells of a piece of research where an audience watching a basketball game on video is told to count each pass.⁵⁵ A person dressed as a gorilla runs onto the court and thumps her hands on her chest and then runs off. The audience members are asked afterward what they remember. Very few individuals mention the gorilla. Even when specifically prompted, they don't recall the gorilla.

What we are paying attention to shapes what we perceive. We manage complexity by applying filters. This comes up in architecture repeatedly. It is why we so emphasize stepping back and bringing in a fresh perspective to help with system concept and architecture validation — and doing this early (while our egos are still relatively uninvested) and then often throughout the life of the architecture. It is vital to be strategic about who participates in these various validation activities. Visual architecting is all about being agile, but we learn important lessons about differentiating value and addressing the architectural challenges through models rather than only through code.

Architecting alone isn't just lonely, it is slower and prone to blind spots. Architecting in big groups is slow too. A small team of two or three, with a lead architect, is often the most agile, working quickly to surface risks and uncertainties. Yes, groupthink can take sway, but a small team that is disciplined about getting external input is less prone to the stranglehold of vested ego. Ego works for and against us. We invest our brain cells into our work — literally! And we deserve to take pride in it. But it can also blind us to weaknesses. So it helps to bring in more eyes and hearts and egos early, to bring fresh perspective and new opportunities, to widen the band of exploration and chaos, and to increase the chances for innovation-yielding eureka's.

And, as we push toward convergence and order in our design, we also need to bring others in to help us assess the efficacy of our approach, lower the risks, and improve our strategies and design. In the Visual Architecting Process we call this “validation,” even though what we are doing is gathering input and ideas as much as we are getting a pat on the back for good progress toward a compelling product and technical strategy. And the pat on the back is important! Positive, enthusiastic response, along with some open-minded questioning, goes a long, long way to stimulating the kind of generative creativity we are looking for from architecture teams. It goes a whole lot further than keeping our architects humble by habitually demonstrating to them that they “don't walk on water.”

Iteration

Our foregone conclusions, our assumptions and assertions, need to be exposed for what they are and tested for validity. Our approach to system design says get some initial stakes in the ground, then validate and move the stakes while it is cheap and quick to move them. Keep doing this. Each additional decision adds to the cost of having to go back and revise earlier decisions, but it is still so much cheaper to do this with system concept and architecture models, and prototypes directed at resolving design issues, than with decisions that are hard-cast in a growing (and all-too-soon massive) code base.

When we talk of iterating in the VAP, we mean repeating the requirements/specification/validation⁵⁶ cycle (see Figure 8), though the content or focus shifts as dominant uncertainties/complexities/challenges are surfaced and addressed, a new round of strategies and decisions are investigated and articulated, and tradeoffs

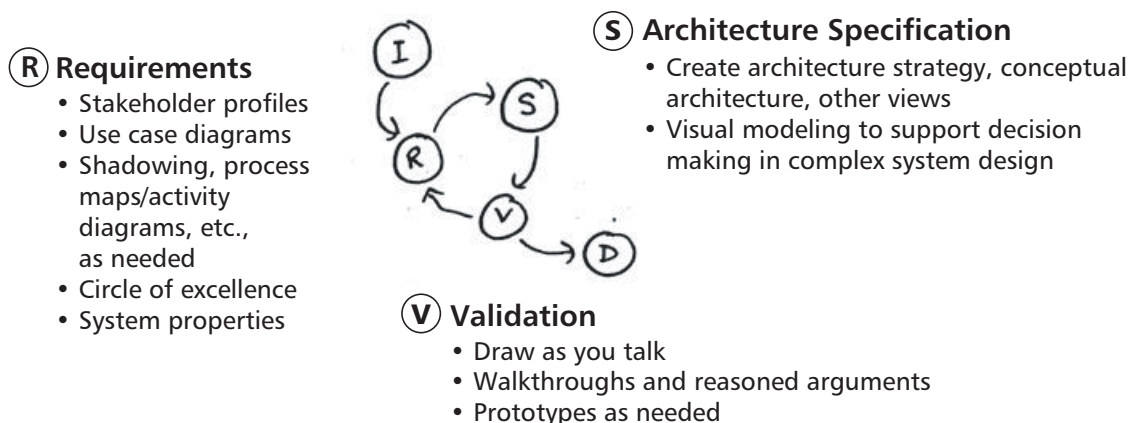


Figure 8 — Requirements, Specification, and Validation phases of the Visual Architecting Process.

are made. It is designedly a learn/improve cycle — that is the whole point of the explicit validation phase of the cycle and the point of iterating. The philosophy is to cycle quickly, and to cycle through different concerns, returning to views to correct, refine, elaborate, refactor, and so on, as more is learned about the requirements and different architectural alternatives are envisaged and explored.

Scope Decisions

Our first milestone (post vision) bringing these exploratory iterative cycles to convergence is the system scoping decision point. System scope shapes what value this system will deliver and how it will support the differentiated identity of the business and its products or services. How we will differentiate dictates what hard problems we will tackle, where we will innovate, and where we must be ahead of the competition. And it allows us to accept “good enough” along those dimensions where we are not trying to create competitive advantage.

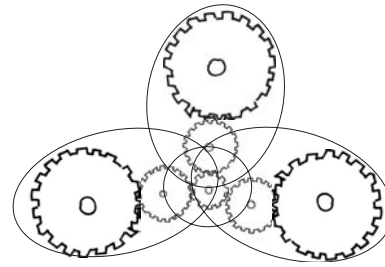
Scoping is about what to out-frame altogether and may point to value partners that need to be found. And it is about what to defer to future releases, what to build in this release, and what it will take to do so.

These decisions are not best made by the architects in isolation, nor by marketing or business or project management — in isolation. They are best made collaboratively, with input from leaders in each of the circles of innovation: customer understanding, business understanding, and technology understanding.

Who leads this consensus process and drives to decisions if the consensus process stalls? This is an organizational decision, but if we want our (lead) architect to be accountable for right system built right, and assuming our architect has the slate of skills to lead across these communities, it makes sense to give the architect end-to-end accountability for both faces of design (user experience and system structure) and empower the architect to lead — remembering that leading is not dictating, but rather inspiring and facilitating dynamic teaming to produce a coherent outcome.

MAKING IT HAPPEN: ORGANIZING FOR INNOVATION

In *The Wheel*, the initial core team (the children) fanned out, and teams of teams formed around them, inspired and aligned by the vision. This was a dynamic, self-organizing process, more organic than we can for the most part achieve given the complexity we deal with. But the symbol of the wheel, the core team a cog around



which other teams turn, is powerful, and it sheds light on scaling innovation processes.

The pace of change is so fast that we heartily want to be agile. But we need to bear in mind that agile means different things depending on complexity. Big projects may be the only way to be agile (quick and responsive to the market) when you want to create something really big. The Empire State building took seven million man-hours to build but was completed in 16 months. That was extreme agile! When we go after big things, we need to get extreme agile on a larger scale than generally talked about in the agile and extreme communities. Yes, extreme agile on that scale takes more upfront architecting to allow the kind of planning it takes to coordinate the effective contribution of more teams, of teams, of teams.

We need to design the cogs — the teams and their interactions — and that design should not be done in isolation from the system architecture or we fall foul of Conway’s Law. To paraphrase, it says that if the design of the system and the design of the organization are at odds, the organization wins — the organization design will drive the system structures, interfaces, and trade-offs among cross-cutting concerns.

With enough of the system architecture thought out, the organization, the teams, can pulse organically because the architecture provides enough design structure to provide rhythm and direction. The architecture will continue to be matured as the system is built out and further innovations are integrated. The architects will need to continue to resolve uncertainty, challenge, and risk, drawing in others as needed to bring expertise and investment to the decisions and to oversee the integrity of the structure as the system evolves and as the teams dynamically adapt in their makeup and their focus.

CONCLUSION

Innovation is increasingly being seen as the antidote to intense global competition, and industry incumbents have to become better at innovating despite their size and inertial histories.

Technology innovations are fundamentally reshaping the landscape of people's lives. The Internet has changed the world and will continue to do so. Software has changed the world even more diffusely and will continue to do so as well. It has changed how people are, how they see themselves, and what they are capable of. Even with just a modicum of artistic ability, we can revel in new outlets for self-expression. Our avenues of exploration and investigation have broadened and deepened. People are connected in ever new ways.

In all this momentum, we need to continue to find new opportunities and new ways to create value and delight and to ignite the passion that fuels customer-to-customer selling. Do we throw up our hands and let small new ventures spark product revolutions, or do we seek to innovate even within industry incumbents? Innovation can come from anywhere: software developers, architects, customers, suppliers, startups — and centenarian giants like GE (number four on *Fast Company's* list of top innovative companies of 2008⁵⁷).

If we shield ourselves with “but,” we yield to the inertial forces that damp innovation. If we change our attitude and begin to wonder, and enroll others in our wondering, we begin to set the wheels in motion that will bring change.

Getting past “but” is about individual attitude and about organizational attitude. Any individual can set the wheels of change in motion and, to quote Ghandi: “Be the change he or she wants to see in the world.” Still, executives set the tone, shaping an organizational attitude that passionately embraces innovation by giving people time to wonder and then to do. The management team needs to create an organizational context that fosters innovation and supports the roles chartered to innovate from concept through design to development and deployment.⁵⁸ Even so, it has to start with individuals being willing to wonder, to dream, and to enroll others in the wondering to create a shared vision picture and then set about doing.

To help adopt this attitude with the courage of conviction, we have shared both an emblematic story and an innovation process that embraces the principles conveyed in the story (summarized in Figure 9). The process speaks to bringing together a team with diverse perspectives to find opportunities to create new, compelling value and to form a shared vision. The story tells us first to dream and then to do. If the staging and work organization is simple and apparent, we can begin to do. Otherwise, we need to do just enough design of the system concept — the user experience that is formed by the web of system capabilities — and just enough

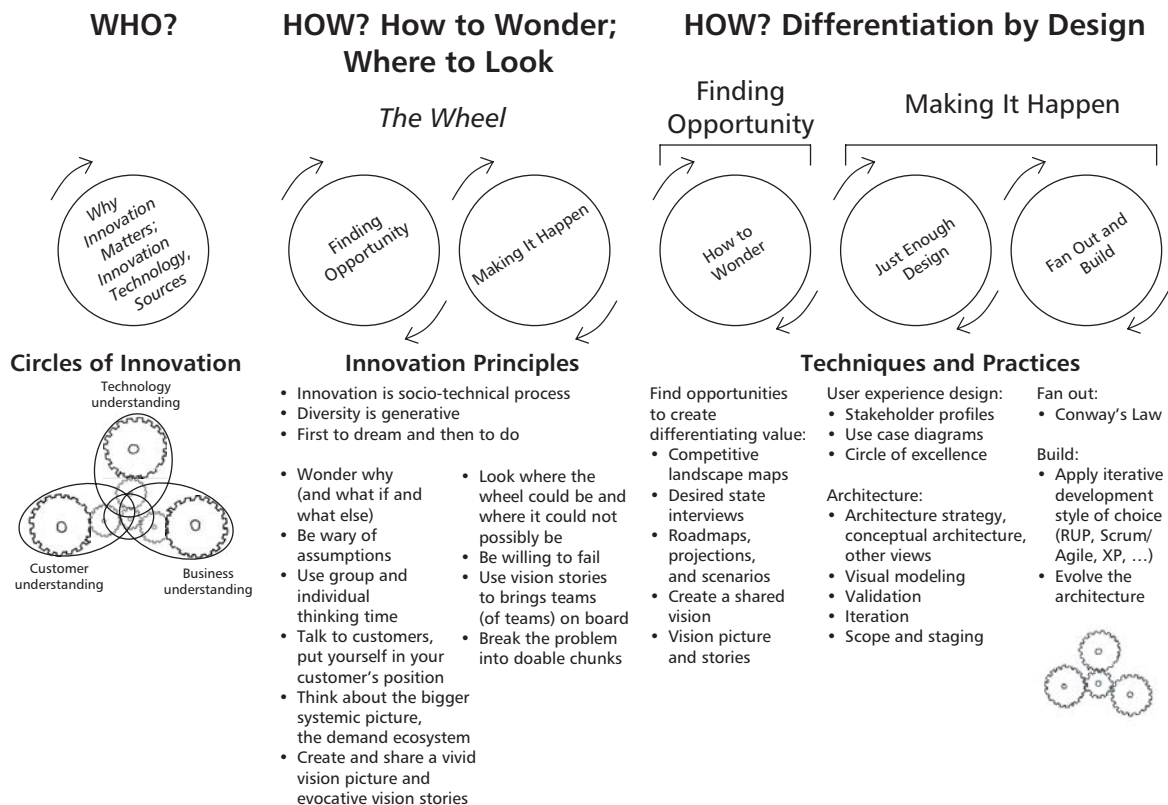


Figure 9 — Summary of the topics in this report.

design of the system structure — the architecture: just enough design to address the dominant uncertainties, challenges, and risks; just enough design to scope the system, decide how to stage development, to design the teams, of teams, of teams that will work together to realize the vision; and just enough design to ensure that the decisions made early won't be the undoing of the system, the strategy it implements, and the business that depends on it. To design is to do, but it is just the beginning. Yet it is the important beginning that sets in motion the cogs, turning cogs, the teams, of teams, working to do, to build the vision, to make it happen.

ACKNOWLEDGMENTS

We would like to express our gratitude to Charlie Alfred, Kristen Sanderson, and Najib Abusalbi whose thoughtful reading and useful feedback improved this report. We would also like to thank Cindy Swain, Kara Letourneau, Lori Goldstein, and Linda Dias at Cutter Consortium. They added the value we would expect from such a competent editorial team, and for that we heartily thank them. But for this report we stretched every rule in the book, and they have been tremendously supportive. We have long been promoting the power of visual modeling and sketching to evoke meaning and make connections through visual image. Demonstrating the value of hand-drawn sketches comes at the expense of polish, and we thank the Cutter team for their support for the messages we wanted to convey both formally through the words of the report, but also informally through our use of hand-drawn sketches.

ENDNOTES

¹Christensen, Clayton M. *The Innovator's Dilemma: When New Technologies Cause Great Firms to Fail*. Harvard Business School Press, 1997.

²Schumpeter, Joseph A. *Capitalism, Socialism and Democracy*. Harper, 1975, originally published 1942.

³Collins, Jim. "The Secret of Enduring Greatness." *Fortune*, 21 April 2008.

⁴Kanter, Rosabeth Moss. "Innovation: The Classic Traps." *Harvard Business Review*, November 2006.

⁵Kim, W. Chan, and Renée Mauborgne. *Blue Ocean Strategy*. Harvard Business School Press, 2005.

⁶Piccoli, Gabriele (ed.). "Starting Off the New Year by Looking Back." *Cutter Benchmark Review*, Vol. 8, No. 1, January 2008.

⁷"Bezos on Innovation." *BusinessWeek*, 17 April 2008 (www.businessweek.com/magazine/content/08_17/b4081064880218.htm).

⁸"Global CEO Study: The Enterprise of the Future." IBM, 5 May 2008 (www.ibm.com/ibm/ideasfromibm/us/ceo/20080505/index.shtml).

⁹Shirky, Clay. *Here Comes Everybody*. Penguin Press, 2008.

¹⁰IBM. See 8.

¹¹Fuller, Buckminster R. *Critical Path*. St. Martin's Griffin, 1982.

¹²Kemper, Steve. *Code Name Ginger: The Story Behind Segway and Dean Kamen's Quest to Invent a New World*. Harvard Business School Press, 2003.

¹³Joachimsthaler, Erich. *Hidden in Plain Sight: How to Find and Execute Your Company's Next Big Growth Strategy*. Harvard Business School Press, 2007.

¹⁴von Hippel, Eric. "Breakthrough Idea Number 6: The HBR List: Breakthrough Ideas for 2007." *Harvard Business Review*, February 2007.

¹⁵Seybold, Patricia. *Outside Innovation: How Your Customers Will Co-Design Your Company's Future*. Collins Business, 2006.

¹⁶Kelley, Tom. *The Ten Faces of Innovation*. Doubleday Business, 2005.

¹⁷Manning, Tony. "Innovate or Die." *Directorship*, June 2000.

¹⁸Bredemeyer, Dana, and Ruth Malan. "What It Takes to Be a Great Enterprise Architect." *Cutter Consortium Enterprise Architecture Executive Report*, Vol. 7, No. 8, August 2004; Malan, Ruth, and Dana Bredemeyer. "Enterprise Architecture as Strategic Differentiator." *Cutter Consortium Enterprise Architecture Executive Report*, Vol. 8, No. 6, June 2005.

¹⁹DeJong, Meindert. *The Wheel on the School*. HarperTrophy, 1972.

²⁰Goldratt, Eliyahu M., and Jeff Cox. *The Goal*. Gower Publishing, 1986.

²¹Hjorth, Daniel (ed.). "What's the Art in the 'Art of Innovation'?" *Cutter IT Journal*, Vol. 21, No. 7, July 2008.

²²Like spokes on a wagon wheel.

²³Kemper. See 12.

²⁴Randy Pausch, author of *The Last Lecture* (Hyperion, 2008) bought a stuffed penguin to give to the first team to fail spectacularly on a class project. That's a great idea, and the sale of stuffed penguins ought to skyrocket — certainly I'll be contributing to this phenomenon! And stuffed storks too! One of our clients has chosen Canadian geese to reinforce teaming as a strong part of its corporate identity, and every employee has a stuffed goose (we were also given two, signaling the extension of their teaming principles to us). Apparently, when it comes to innovation, water birds are happening!

²⁵Now, you could object this is not about innovation at all; since storks nest on wheels on roofs in other Dutch towns and even did nest on roofs in Shora long ago, you could say this is just a story about children using a known solution to a known problem. But innovation is not invention, at least not necessarily; innovation may be the application of a known approach in a new domain, or a new combination of known approaches. Some of the sources of innovation ideas are, in

Erich Joachimsthaler's words, "hidden in plain sight" (see end-note 13). Innovation is a lot about transferring what is known to a problem, which in its context is new, and sometimes the problem is simply a problem that has to be recast, so that known solutions become applicable to it.

²⁶Malan and Bredemeyer, 2005. See 18.

²⁷The Pareto principle, also known as the 80-20 rule, states that for many events, 80% of the effects come from 20% of the causes.

²⁸The Visual Architecting Process is described in Chapter 2 of our draft *Software Architecture Action Guide* (www.bredemeyer.com/ArchitectingProcess/SWAAActionGuideTOC.htm).

²⁹Malan and Bredemeyer, 2005. See 18.

³⁰Joachimsthaler. See 13.

³¹Ware, Colin. *Visual Thinking for Design*. Morgan Kaufmann, 2008.

³²Russell L. Ackoff, author of *Idealized Design* (Wharton School Publishing, 2006), argues against time traveling into the future to envision our desired state. He contends we need to formulate the state we want now. Ackoff argues (much like many agilists) that we aren't good at predicting the future and concludes we shouldn't design "in" the future. I think it is a matter of how Ackoff is framing up this "going to the future." If we keep ourselves grounded in what we know and can find out about (potential) customer needs and what is happening around us in terms of technology capability and user expectations, we're not flying above the ozone to apply evident trends and patterns. We have found that asking people to think of "being" in the time frame of the system delivery, rather than now, frees them up from that obstacle-focused rut we tend to be so stuck in. But if our stakeholders have Ackoff's orientation to "time traveling," don't call it that! Don't let the name of a thing get in the way of what is important.

³³Malan, Ruth, and Dana Bredemeyer. "Creating an Architectural Vision: Collecting Input," 25 July 2000 (www.bredemeyer.com/pdf_files/vision_input.pdf).

³⁴Coplien, James, and Neil Harrison. *Organizational Patterns of Agile Software Development*. Prentice Hall, 2004.

³⁵Rechtin, Eberhardt. *Systems Architecting*. Prentice Hall, 1991.

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³⁷Bazerman, Max H., and Michael D. Watkins. "Predictable Surprises: The Disasters You Should Have Seen Coming." *Harvard Business Review*, March 2003.

³⁸Bazerman and Watkins. See 37.

³⁹The prefrontal cortex distinguishes us from (other) animals, giving us the ability to envision a future beyond the immediate.

⁴⁰Behar, Howard. *It's Not About the Coffee: Leadership Principles from a Life at Starbucks*. Penguin Group, 2007.

⁴¹We can't go into every wrinkle in the process, but it is fair to ask, "What if the team is not aligned? What if the team sees

different opportunities and can't converge on a choice of identity and value propositions to deliver?" William Shatner gave a talk at the June 2008 *IBM Rational Software Development Conference*, in which he said given that no one can know which idea is a market winner, "everybody's opinion has an equal value with everyone else's," hence organization matters and having a designated leader matters. We will be called upon to compromise to save the vision, and it begs the questions, "When do you stand on principle and when do you compromise? When are you obstinate and when are you amenable?" People's opinions count, but at a certain point there has to be a leader who makes the decision, who says "I've listened, and this is the way we'll go." This is sage advice and very salient to system development, not just movie direction.

⁴²We highly recommend Stephen Denning's book, *The Leader's Guide to Storytelling: Mastering the Art and Discipline of Business Narrative* (Jossey-Bass, 2005). There are just too many points and all of them beg to be made in this report! Another worthwhile reference is an article by Robert McKee ("Happy Tales: The CEO as Storyteller," *Harvard Business Review*, June 2003), in which he advises us to tell the tale of "the struggle between expectation and reality in all its nastiness," to include the inciting incident when the day-to-day balance is disturbed; the subjective expectation that life will continue in balance; and the notion that objective reality is uncooperative and full of antagonists and opposing forces that create real struggles with which the protagonist has to contend.

⁴³Chouinard, Yvon. *Let My People Go Surfing: The Education of a Reluctant Businessman*. Penguin Press, 2005.

⁴⁴Rechtin. See 35.

⁴⁵Roam, Dan. *The Back of a Napkin*. Portfolio Hardcover, 2008.

⁴⁶As warranted by the complexity of the system or development environment.

⁴⁷Some might say functional design, but we specifically mean designing system capabilities that are part function ("what" or service) and part property ("how well" or service level) as well as the warp and woof that weaves the capabilities together into the full tapestry of the customer experience.

⁴⁸Scoping decisions involve deciding on system capabilities, timetables, and resources. As soon as we set time to market and budget, however, we constrain the achievable system scope (capabilities). This is known as the "iron triangle," and, given the mythical man-month in software development, it means we will run out of time if we scope unrealistically.

⁴⁹These terms reference our Architecture Decision Model, which is described in Chapter 1 of our draft *Software Architecture Action Guide* (www.bredemeyer.com/ArchitectingProcess/SWAAActionGuideTOC.htm).

⁵⁰Charlie Alfred is doing important work in this area. Charlie blogs at <http://charliealfred.wordpress.com>.

⁵¹Some suggest that it is an unnecessary overload to model twice, once in architecture diagrams and again in the code. Yes, it does mean we have to keep the architecture models

(and supporting explanations, rationale, etc.) up to date and that takes conscious additional effort, which is hard to motivate since code is what our organizations typically value and measure progress on. But it is not duplication of effort. Architecture is used to orchestrate the social work of building and evolving complex systems, and architecture is used to tackle hard design challenges and make tradeoffs while it is relatively cheap to play out various alternatives and assess their relative efficacy. Architecture is architecture. It is good for what it is good for: thinking about, designing, and communicating the overall structure of the system and the important architectural mechanisms.

Architecture models and documents play many roles. They help the architects think about the many, many challenges they have to balance and address. They help the architects get input, create a shared mind-space where others can play a role in seeing the alternatives being considered, pose new approaches, throw challenges at them, and advance the understanding of the system and what it must do. And the models document decisions, so these decisions can be explained and shared. And they create a record of decisions, so these decisions can be referred to (and defended and amended if necessary) during all phases of the life of the system. How much documentation you need is context-dependent. Tomorrow can take care of itself better if it is clear what we did today! Code as architecture, code as model of the system, just does not scale as a communication tool when you're trying to build and evolve 100,000 to million++ LOC systems.

⁵²The ESD Architecture Committee. "The Influence of Architecture in Engineering Systems." *MIT Engineering Systems Monograph*, March 2004.

⁵³Origami is an age-old craft, but it was revolutionized by the creation of a design language. As quoted by Robert Lang on the Apple site: "The modern art form was born in the 20th century when a Japanese artist named Akira Yoshizawa started creating new figures of artistic beauty that inspired other origami artists to expand their horizons." Yoshizawa also created an instructional language that then enabled them to share their creations and build off of each others' work" (www.apple.com/science/profiles/origami). Now complexity that would have been incomprehensible is manageable, and ever more innovative designs are being created to astonish and amaze us.

Software design languages like UML have also played their part in making design advances, and our field has seen dramatic increases in system complexity, in part because the design language gives designers a means to visualize, explore, and communicate design elements (separating design decisions from implementation details), and in part because design knowledge can be captured and shared, in particular in the form of patterns. In software, patterns have been formalized to mean not just the design element in structural and behavioral model form, but also a clear characterization of the problem being addressed and the context in which the solution is a proven good fit to the problem.

⁵⁴Rabe, Cynthia Barton. *The Innovation Killer: How What We Know Limits What We Can Imagine — and What Smart Companies Are Doing About It*. AMACOM, 2006, p. 39.

⁵⁵Grandin, Temple. *Animals in Translation*. Scribner 2004.

⁵⁶For more on iterating through the Architecture Decision Model (strategy, conceptual, logical, and execution architecture), applying VAP requirements/structuring/validation cycles, we refer you to our Web site (www.bredemeyer.com) and the draft *Software Architecture Action Guide* book.

⁵⁷"The World's Most Innovative Companies." *Fast Company* (www.fastcompany.com/magazine/123/the-worlds-most-innovative-companies.html).

⁵⁸Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I shall move the world." As Charlie Alfred has pointed out, the position of the fulcrum is crucial: if it is too far from the object and too close to your hands, you can't use the lever to move anything. Architects who are chartered to lead from system concept formulation to system delivery have the fulcrum positioned to succeed. Architects who are expected to take a requirements document fait accompli and create a design have little leverage. We cannot cover every caveat in this report, but we hope to follow up with a book to deal more expansively with these and other critical topics.

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